Q1) ETL Operation Using Amazon Data Glue

Part (a): Create S3 Bucket and Folder Structure

Step 1: Navigate to Amazon S3 Log in to the AWS Management Console. In the search bar, type **S3** and click on it. You will land on the S3 Buckets dashboard.

Step 2: Create a New S3 Bucket Click **Create bucket**. Fill in the following details:

- Bucket Name: company-etl-glue-bucket
- AWS Region: us-east-1
- Block all public access: Enabled
- Bucket Versioning: Disabled
- Encryption: SSE-S3

Click **Create bucket**.

Step 3: Create Folder Structure Open the bucket and create the following folders:

- raw-data/employee/ — for employee CSV
- raw-data/department/ — for department CSV
- processed-data/joined/ — for Glue job output
- processed-data/temp/ — for temporary Glue files

Step 4: Upload CSV Files Navigate to raw-data/employee/, click **Upload**, and upload employee.csv. Repeat for raw-data/department/ with department.csv.

Sample employee.csv columns: emp_id, name, age, salary, dept_id Sample department.csv columns: dept_id, dept_name, location

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Part (b): Create Glue Database, Catalog, and Crawlers

Step 1: Open AWS Glue In the AWS Console search bar, type **Glue** and open AWS Glue.

Step 2: Create a Glue Database Go to **Data Catalog → Databases → Add database**.

- Database Name: company_etl_db
- Description: Database for Employee and Department ETL pipeline

Click **Create database**.

Step 3: Create IAM Role for Glue Go to **IAM → Roles → Create role**. Select **AWS Service → Glue**. Attach these policies:

- AWSGlueServiceRole
- AmazonS3FullAccess

Name the role AWSGlueServiceRole-ETL and click **Create role**.

Step 4: Create Crawler for Employee Data Go to **Glue → Crawlers → Create crawler**.

- Name: crawler-employee

- Data source: S3 path → s3://company-etl-glue-bucket/raw-data/employee/
- IAM Role: AWSGlueServiceRole-ETL
- Target database: company_etl_db
- Table name prefix: raw_

Click **Create crawler**.

Step 5: Create Crawler for Department Data Repeat above with:

- Name: crawler-department
- S3 path: s3://company-etl-glue-bucket/raw-data/department/

Click **Create crawler**.

Step 6: Run Both Crawlers Select both crawlers and click **Run**. Wait for status to show **Succeeded**. Go to **Data Catalog → Tables** — you should see raw_employee and raw_department tables created automatically.

Part (c): Add Transforms — Null Removal, Timestamp, and Join

Step 1: Open AWS Glue Studio In Glue, go to **ETL Jobs → Create job → Visual ETL**.

Step 2: Add Data Sources Add two **AWS Glue Data Catalog** source nodes:

- Source 1: Database company_etl_db, Table raw_employee
- Source 2: Database company_etl_db, Table raw_department

Step 3: Remove Null Values from Employee Table Connect a **Drop Null Fields** transform to the raw_employee source. Configure it to drop rows where emp_id, name, or dept_id is null.

Step 4: Remove Null Values from Department Table Connect a **Drop Null Fields** transform to the raw_department source. Configure it to filter rows where dept_id or dept_name is null.

Step 5: Add Current Timestamp Column Connect a **Derived Column** (Add Columns) transform after the employee null-filter node.

- New Column Name: processed_at
- Expression: current_timestamp()
- Data Type: timestamp

Step 6: Join Employee and Department Tables Add a **Join** transform node. Connect both the timestamp-added employee node and the null-filtered department node as inputs.

- Join Type: Inner Join
- Left Key: dept_id (from Employee)
- Right Key: dept_id (from Department)

Part (d): Configure Destination and Run Glue Job

Step 1: Add S3 Destination After the Join node, add a **Data Target → S3** node.

- Format: Parquet
- Compression: Snappy
- S3 Target Path: s3://company-etl-glue-bucket/processed-data/joined/
- Update Data Catalog: Enabled

Step 2: Configure Job Details Click the **Job details** tab.

- Job Name: etl-employee-department-join
- IAM Role: AWSGlueServiceRole-ETL
- Glue Version: Glue 4.0
- Worker Type: G.1X
- Number of Workers: 2
- Temp Directory: s3://company-etl-glue-bucket/processed-data/temp/

Step 3: Save and Run the Job Click **Save**, then click **Run**. Go to the **Runs** tab to monitor. Wait for status to show **Succeeded**.

Step 4: Verify Output in S3 Go to S3 → company-etl-glue-bucket → processed-data/joined/. You should see Parquet output files (part-00000-*.parquet) generated by the Glue job.

Summary Table

Stage	Component	Details
Extract	Amazon S3	employee.csv, department.csv in raw-data/
Extract	AWS Glue Crawlers	crawler-employee, crawler-department
Extract	Glue Data Catalog	company_etl_db with raw_employee, raw_department tables
Transform	Drop Null Fields	Filters null rows from both tables
Transform	Add Timestamp	processed_at = current_timestamp()
Transform	Join	Inner join on dept_id
Load	S3 Destination	processed-data/joined/ in Parquet format
Load	Glue Job Run	etl-employee-department-join: Succeeded